Chapter 3 Notes

3.1

In the proof of Theorem 3.1, here is why the class of all transitive natural numbers is a set. We can show that it is a subset of ω, and is hence a set by A_2. For suppose x is a member of the class of all transitive natural numbers. Then obviously x is also a natural number, hence $x \in \omega$, thus completing the proof.

Note that here is why the only natural numbers allowed for by Definition 2.2 are the ones defined right before Definition 1.1. By Definition 2.2, a natural number is a member of every inductive set. Now, consider the following inductive set: $\emptyset$ is an element, and if x is an element, then so is x^+. Hence every natural number is a member of this set. But this set is just the set of all sets that were defined in the way just before Definition 1.1.

Exercise 3.1

Suppose that A is a class of natural numbers such that for every natural number n, if A contains all numbers less than n, then A contains n as well. Now, we can prove by induction that A contains all natural numbers. Consider the property P meaning A contains all numbers natural numbers less than or equal to n. Now, there are no natural numbers less than 0. Thus the statement that A contains all natural numbers less than 0 is vacuously true; hence A contains 0. Thus, because 0 is the only natural number less than or equal to 0, P holds for 0. Now suppose that P holds for some n, that is, A contains all natural numbers less than or equal to n. But then A contains all numbers less than $n+1$, meaning (by the definition of A) A also contains $n+1$. Thus, by induction, A contains all natural numbers.

Exercise 3.2

Consider a natural number n.

(a): Suppose that $x \in n$. Now, it cannot be that $n = \emptyset$. Thus n is non-empty, hence of the form $(n-1)^+$. Thus either $x \in n-1$ or $x = n-1$. Now, $\cup n$ is just the set of all elements of all elements of $n-1$ together with all elements of $n-1$. Now suppose $x = n-1$. Then $x \notin \cup n$. For suppose it were. Then it would either be an element of $n-1$ or it would be an element of an element of $n-1$. In the first case, we would have $n-1 \in n-1$, which is ruled out by Theorem 3.2. In the second case, there would be some y such that $x \in y$ and $y \in n-1$. But then, by the transitivity of $n-1$ (by Theorem 3.1), $x \in n-1$, that is, $n-1 \in n-1$, which is again ruled out by Theorem 3.2. Thus $n-1 \notin \cup n$, meaning the statement is not necessarily true.

It should be noted that all elements other than $n-1$ of n are members of $\cup n$. For all elements of n

are either elements of $n-1$ or equal to $n-1$. Thus, if they are not equal to $n-1$, the element is an element of $n-1$. But, because $n-1 \in n$, then all elements of $n-1$ are elements of $\cup n$.

(b): Suppose that $x \in \cup n$. Thus either $x \in n-1$ or x is an element of some element of $n-1$. In the first case, because $n-1 \in n$, $x \in n$ (by the transitivity of n). In the second case, because $n-1$ is transitive, it follows that $x \in n-1$. But then, because $n-1 \in n$ and n is transitive, $x \in n$. Thus, in either case, $x \in n$, and thus $\cup n \subseteq n$. Hence the statement is necessarily true.

(c): By part (b) of this problem, it follows that $x \in \cup n$ implies $x \in n$. But $n^+ = n \cup \{n\}$. Thus all elements of n are in n^+, meaning $x \in n^+$. On the other hand, suppose that $x \in n^+$. Then it could be that $x = n$. But then, if $x \in \cup n$, we would have either n being an element of an element of $n-1$, or $n \in n-1$. In the first case, the transitivity of $n-1$ would show that $n \in n-1$. Thus we would have, in either case, $n \in n-1$, which is impossible. For if $n \in n-1$, then, because $n-1 \in n$, $n-1 \in n-1$ by the transitivity of $n-1$, which is impossible (or, if $n-1 = \emptyset$, it could have no members). Hence it is not necessarily the case that $n^+ \subseteq \cup n$. Thus $\cup n = n^+$ does not necessarily hold.

Similar reasoning as to the above also shows that $n-1$ cannot be an element of $\cup n$. It should be noted that $n-1$ and n are the only members of n^+ that are not members of $\cup n$. For any other element of n^+ is an element of n but not equal to $n-1$. But $n = n-1 \cup \{n-1\}$. Hence it would be an element of $n-1$, and because $n-1$ is an element of n, it would also be an element of $\cup n$.

(d): On the one hand, suppose that $x \in \cup n^+$. Then either x is an element of an element of n, or x is an element of n. In the first case, because n is transitive, $x \in n$. The second case means $x \in n$. Thus, in any case, $x \in n$. On the other hand, suppose that $x \in n$. Then $x \in \cup n^+$, as $n^+ = n \cup \{n\}$. Hence $\cup n^+ = n$.

(e): By part (b) of this problem, $\cup n \subseteq n$ holds. But by the answer to part (a) of this problem, it does not necessarily follow that $n \subseteq \cup n$. Thus it is not necessarily true that $\cup n = n$.

Exercise 3.3

(a): On the one hand, suppose that $x \in \cup\omega$. Then x is an element of a natural number. Thus either $x \in n$ or $x = n$. Now we prove the following statement: every element of every natural number is a natural number. The base case vacuously holds, as $\emptyset$ has no elements. Now suppose the statement holds for some n. Then $n^+ = n \cup \{n\}$. Hence, if $y \in n^+$, either $y \in n$ or $y = n$. In the former case, by the induction assumption, it follows that y is a natural number, and in the latter case, y would be a natural number (as n is a natural number). Thus, in any case, y is a natural number, therefore completing the proof. As a consequence of having proven this statement, it follows that x is a natural number, and hence $x \in \omega$. On the other hand, suppose that $x \in \omega$. Thus, for some n, $x = n$. Now, all elements of n^+ are in $\cup\omega$. But n is an element of n^+. Thus $n \in \cup\omega$, and hence $x \in \cup\omega$. Thus we have proven that $\cup\omega = \omega$.

(b): On the one hand, suppose that $x \in \cup\omega^+$. Now, $\omega^+ = \omega \cup \{\omega\}$. Thus either x is an element of a natural number or x is a natural number. By the results of part (a) of this problem, it follows that if x is an element of a natural number, then x is a natural number. Thus x is a natural number, that is, $x \in \omega$. Now suppose that $x \in \omega$. Then $x \in \cup\omega^+$, as x would be an element of ω, which is an element of ω^+. Thus we have proven that $\cup\omega^+ = \omega$.

Note that a generalization cannot be made of this problem. Define $\{\omega_0, \omega_1, \omega_2, ..., \omega_n, ...\}$ in the following way: $\omega_0 = \omega$ and $\omega_{n+1} = \omega_n^+$. Then it does not follow that $\cup\omega_n = \omega$ for all n. For consider $n = 2$. This is ω_1^+, that is,

$$\omega_1 \cup \{\omega_1\} = (\omega \cup \{\omega\}) \cup \{(\omega \cup \{\omega\})\}$$

Thus $\omega \in \cup\omega_1^+$. But $\omega \notin \omega$, meaning $\cup\omega_2 \neq \omega$. Indeed, it is a general fact that, for $n \geq 2$, $\cup\omega_n \neq \omega$. To prove this, we can just prove that, for all $n \geq 2$, $\omega \in \omega_n$. We have just proven the base case. If the result

holds for some n, then $\omega \in \omega_{n+1}$, as $\omega_{n+1} = \omega_n \cup \{\omega_n\}$, and it was assumed that $\omega \in \omega_n$. Thus, for every $n \geq 2$, $\omega \in \omega_n$. Now we can prove the statement that, for $n \geq 2$, $\cup\omega_n \neq \omega$. We proved the base case for $n = 2$ above. Now consider some $n > 2$. Then $\omega_n = \omega_{n-1} \cup \{\omega_{n-1}\}$ and $\cup\omega_n$ consists of all elements of all elements of ω_{n-1} and all elements of ω_{n-1}. Because ω is an element of ω_{n-1}, it follows that $\omega \in \cup\omega_n$, and thus, because $\omega \notin \omega$, $\cup\omega_n \neq \omega$.

3.4

Consider an aritrary class A and an arbitrary function g. We shall say that A is inductive under g, or g-inductive, if the following two conditions hold:
(1) $\emptyset \in A$,
(2) for every x in A, the set $g(x)$ is in A.
We shall say that A is minimally inductive under g if A is inductive under g and if no proper subclass of A is inductive under g, in other words, any subclass of A that is inductive under g contains all elements of A.
Supppose M is minimally inductive under g and that R is a relation satisfying the following two conditions:
D_1 $R(x, 0)$ holds for every x in M;
D_2 for all x, y in M, if $R(x, y)$ and $R(y, x)$ then $R(x, g(y))$;
Then $R(x, y)$ holds for all x, y in M

In Smullyan's proof, we calls an element x of M left normal if $R(x, y)$ holds for all y in M, and right normal if $R(y, x)$ holds for all y in M. Then he shows that every element that is right normal is also left normal. After this, he shows that all elements are right normal (and hence also left normal), and thus $R(x, y)$ holds for all x, y in M. But I do not think we need to show that every right normal element is also left normal. For suppose we have shown that all elements are right normal. Thus, for all x in M, $R(y, x)$ holds for all y in M. Now consider any elements x, y in M. Then y is right normal, meaning $R(x, y)$ holds.